

PASTIS: agentic scientific text simplification

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TURN IT SIMPLE



CLEF SimpleText Task 1

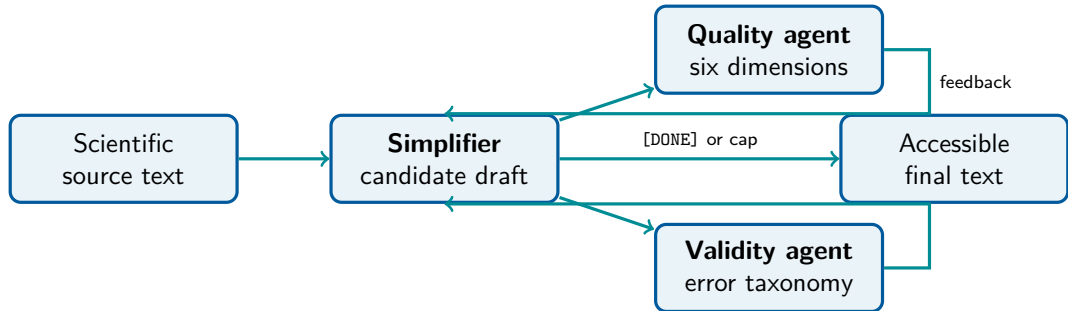
Generate accessible versions of scientific texts for non-specialist readers.

- **Subtask 1.1:** simplify individual sentences
- **Subtask 1.2:** simplify complete abstracts
- **Data:** biomedical systematic reviews paired with lay summaries

Research question

Can iterative agents improve readability while preserving the source meaning?

Approach: prompt-based
Fine-tuning: none



The agents iterate until the Simplifier terminates or the iteration budget is exhausted.

The two critics optimize complementary properties

Quality agent

- Lexical simplicity
- Syntactic simplicity
- Cohesion
- Concept simplicity
- Discourse coherence
- Personalization

Validity agent

- Fluency errors
- Alignment: Format or prompt misalignment
- Information: Hallucinations and topic shift
- Simplification: Overgeneralization, overspecification, Lost or out-of-scope information

Target: CEFR B1–B2 vocabulary and approximately grade 8–9 readability.

The system runs locally without fine-tuning

Component	Configuration
Base model	Gemma 4 (E4B), open-weight
Hardware	2 × RTX 4090 GPUs
Inference	vLLM, batched
Output control	constrained JSON (xgrammar)
Decoding	temperature = 0
Fine-tuning	none

Refinement budgets

8 iterations
sentence level

20 iterations
document level

Average self-termination: **2.2 iterations** for sentences and **4.7 iterations** for abstracts.

Run	SARI ↑	BLEU ↑	FKGL ↓
Source baseline	8.88	12.40	13.02
Best overall system	47.43	14.21	10.19
Best open-weight system	47.01	11.97	10.55
PASTIS	44.70	5.21	7.59

Competitive overlap

SARI is close to the strongest systems and around the median of 127 runs.

Strong readability gain

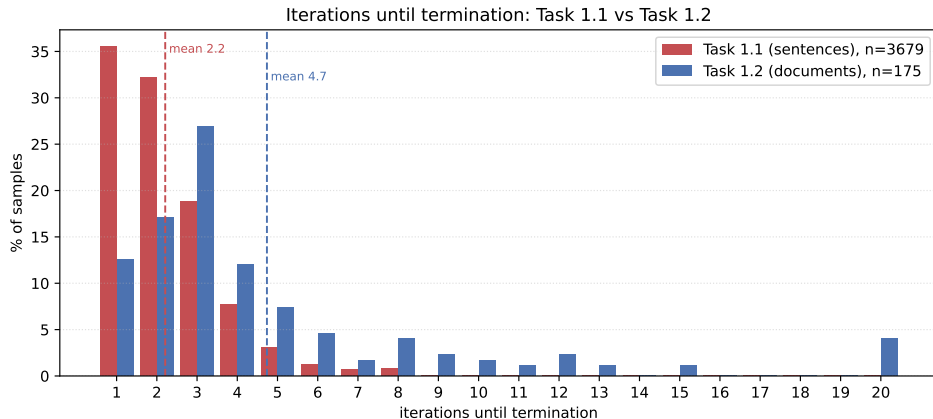
FKGL falls by **5.43 grade levels** relative to the source.

Document-level results show the same trade-off

System	Approach	SARI	BLEU	FKGL
deepseek-v3	prompting	48.85	12.22	8.80
RAG + best-of- N	retrieval	47.57	12.80	12.34
Gemma-2B-IT	QLoRA	47.18	10.42	9.91
Gemma-4	prompting	46.22	9.69	7.77
PASTIS	agentic	44.25	6.84	9.53
MedGemma	domain LLM	43.84	7.92	8.06
BARTsent	fine-tuned seq2seq	38.97	6.97	12.01
Source	baseline	8.88	12.40	13.02

PASTIS reaches median-level SARI while reducing estimated reading difficulty by **3.49 grade levels**.

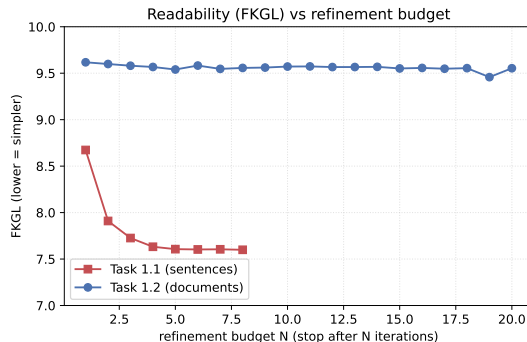
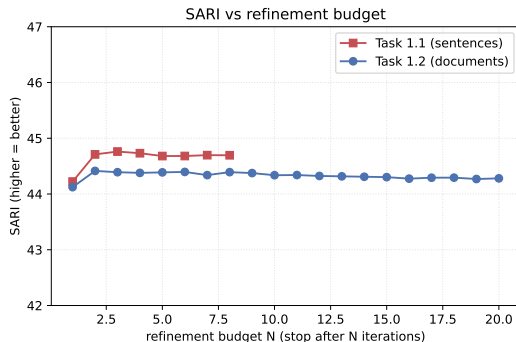
Sentences converge faster than abstracts



Mean iterations: **2.2** for sentences vs. **4.7** for abstracts; medians: 2 vs. 3.

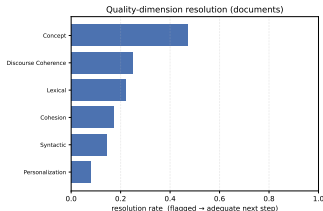
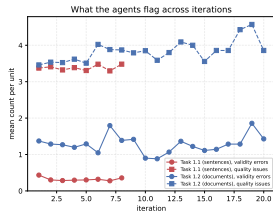
Iteration improves readability, but barely moves SARI

Effect of iterative refinement (official doc-aggregated scoring)



SARI peaks early; document readability gains are front-loaded, whereas sentence FKGL continues to fall.

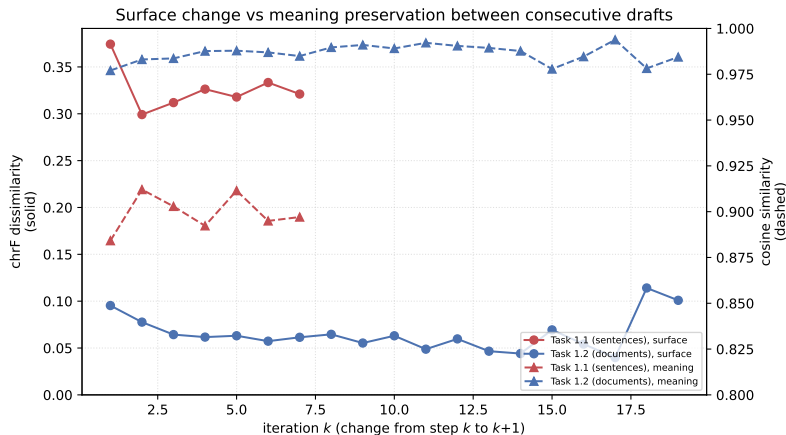
The loop keeps editing rather than settling



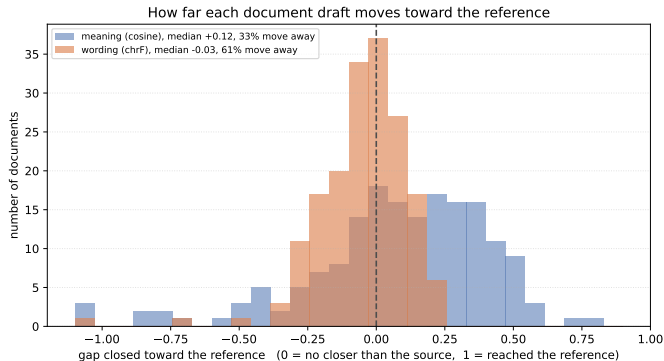
- Quality flags stay near 3.4 per sentence and 3.5–4 per abstract.
- About 18% of flagged dimensions become adequate at the next step.
- A similar number of new issues appears after rewriting.
- Validity errors decline modestly for documents: 1.37 → 1.05.

The system reaches a **churn equilibrium**: each rewrite resolves issues but also exposes new ones.

Documents preserve meaning; sentences undergo larger edits



- Document drafts remain semantically close across steps (cosine 0.98–0.99).
- Sentence drafts show larger surface changes and more semantic drift (cosine 0.88–0.91).



Semantic distance

Median gap closed: **+0.12**; final drafts move slightly toward the reference meaning.

Surface distance

Median gap closed: **-0.03**; 61% move away from the reference wording.

- 1 **An open-weight, local, prompt-based system is competitive.** PASTIS reaches SARI 44.70 at sentence level and 44.25 at document level without fine-tuning.
- 2 **The main measurable gain is readability.** FKGL improves from 13.02 to 7.59 for sentences and 9.53 for abstracts.
- 3 **Most overlap performance is established in the first draft.** Additional refinement changes SARI by less than one point.
- 4 **Single-reference metrics miss useful reformulation.** The loop preserves or approaches meaning while using wording different from the gold reference.

Limitations

- Single-reference evaluation
- Biomedical domain only
- Limited system-level ablation
- Some sentence-level semantic drift

Next steps

- Evaluate with multiple references and humans
- Improve stopping criteria
- Reduce quality-flag churn
- Optimize readability and faithfulness jointly



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Thank you

Questions?

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This research was funded by the French National Research Agency (ANR),
ref. ANR-22-CE23-0019-01.